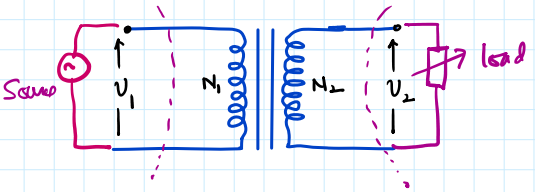
Lecture 36: Transformer: Electrical Circuit

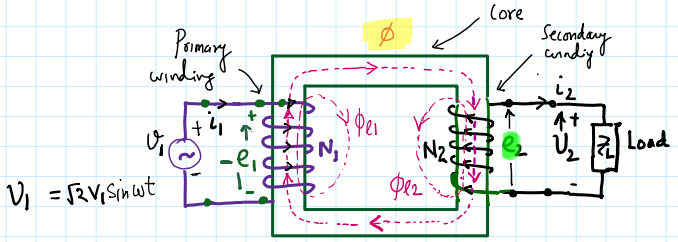
Principle of Operation of Transformer and its analysis:

Primary Secondary



Components of Transformer:

- ✓ Magnetic core
  - Soft iron
  - Laminated silicon steel
  - Ferrite core
  - Amorphous core
  - CRGO (Cold reinforced grain oriented steel)
- ✓ Primary winding
- ✓ Secondary winding

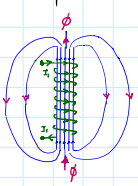


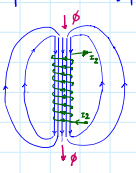
Clockwise Ampere Turns creates flux upward into the plane

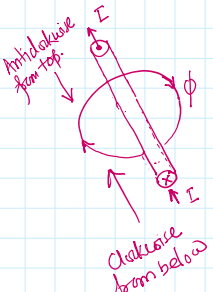
Anticlockwise Ampere-Turns due to downward flux lines out of plane

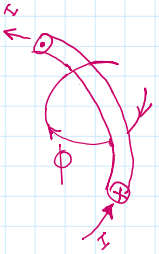


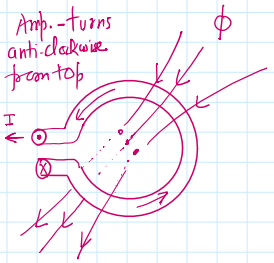
Right hand rule











Ampere-Turns clockwise from below the plane

$V_1 \rightarrow e_1 = -N_1 \frac{d\phi}{dt} = -L_1 \frac{di_1}{dt}$

Schematic representation of transformer

linear relationship between i & ϕ $\Rightarrow$ $L_1 i_1 = N_1 \phi$

Ideal Transformer: Two assumptions

1) No losses in transformer
2) Infinite permeability of core material

$V_1 \rightarrow i_1 \rightarrow \phi$; $L_1 i_1 = N_1 \phi \Rightarrow \boxed{i_1 = \frac{N_1}{L_1} * \phi}$ (i_ϕ)

$\phi = \phi_{max} \sin \omega t$; $\bar{\phi} = \frac{\phi_{max}}{\sqrt{2}} \angle 0^\circ$

$e_1 = -N_1 \frac{d\phi}{dt} = -N_1 * \omega \phi_{max} \cos \omega t$

$$e_1 = -N_1 \frac{d\phi}{dt} = -N_1 * \omega \phi_{max} \cos\omega t$$

$$\cos\omega t \rightarrow \sin\left(\frac{\pi}{2} - \omega t\right) \rightarrow -\sin(\omega t - \pi/2)$$

$$e_1 = N_1 \omega \phi_{max} \sin(\omega t - \pi/2)$$

$$\bar{E}_1 = \frac{N_1 \omega \phi_{max}}{\sqrt{2}} \angle -90^\circ \quad \text{(1)}$$

$$e_2 = -N_2 \frac{d\phi}{dt} = -N_2 \omega \phi_{max} \cos\omega t = N_2 \omega \phi_{max} \sin(\omega t - \pi/2)$$

$$\Rightarrow \bar{E}_2 = \frac{N_2 \omega \phi_{max}}{\sqrt{2}} \angle -90^\circ \quad \text{(2)}$$

$$\frac{E_1}{E_2} = \frac{N_1}{N_2} = \text{Transformer Ratio} = a = \text{Turns Ration}$$

$$\downarrow$$

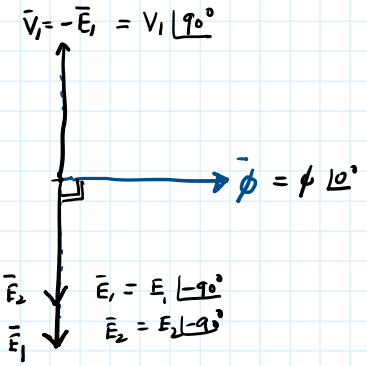
$$V_1 \simeq E_1$$

$$V_2 \simeq E_2$$

$$\frac{V_1}{V_2} = \frac{E_1}{E_2} = \frac{N_1}{N_2} = a = \frac{I_2}{I_1}$$

$$V_1 I_1 = V_2 I_2$$

$$\frac{V_1}{V_2} = \frac{I_2}{I_1} = \frac{N_1}{N_2} = a$$



Phasor diagram at no load and lossless transformer

$$E_1 = \frac{N_1 \omega \phi_{max}}{\sqrt{2}} = \frac{2\pi f N_1 \phi_{max}}{\sqrt{2}} = \sqrt{2}\pi f N_1 \phi_{max}$$

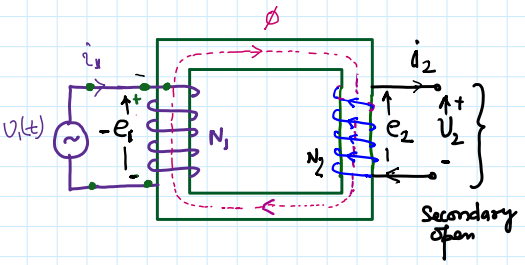
$$E_1 = 4.44 f N_1 \phi_{max}$$

$$E_2 = 4.44 f N_2 \phi_{max}$$

$$\frac{E_1}{E_2} = \frac{N_1}{N_2}$$

Transformer on NO LOAD:

At no load, the load is not connected hence $i_2 = 0$. However some input current flow to setup magnetic flux in the core. This flux magnetises and demagnetizes core at supply frequency, hence causes core losses.



These core losses or iron losses are divided into two losses.

1) Hysteresis current losses (P_h)

2) Eddy current losses (P_e)

Thus $P_c = (P_h + P_e) * \text{Volume of Core}$

Through an empirical relation, it is found that

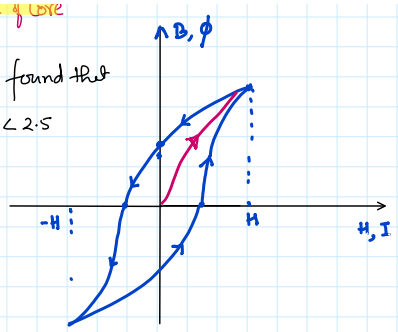


Thus $P_c = (P_h + P_e) * \text{Volume of core}$

Through an empirical relation, it is found that

$P_h = K_h * f * B_{max}^{x}$; $1.5 < x < 2.5$

$P_e = K_e f^2 B_{max}^2$



K_h is proportionality constant depends upon, volume and quality of core

K_e is proportionality constant that depends upon volume, resistivity and thick of lamination

B_{max} = Maximum flux density

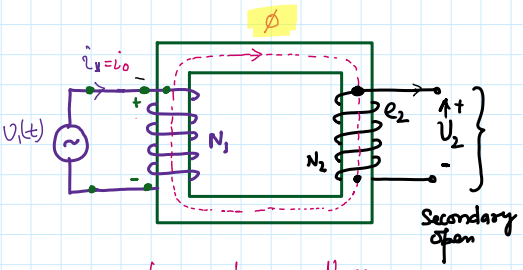
f = frequency of alternating flux

At no load, no load current i_o flows to account the core losses and magnetizing component of current to setup magnetic flux in the core, i.e.

$$i_1 = i_0 = i_c + i_\phi$$

$$\bar{I}_0 = \bar{I}_c + \bar{I}_\phi$$

↑ accounts for the core losses



Transformer at no voltage

$$\bar{S} = \bar{S}_1 = \bar{V}_1 \bar{I}_0^* = \bar{V}_1 (I_0 \angle -\phi_0)$$

$$= V_1 I_0 (\cos\phi_0 + j \sin\phi_0)$$

$$= V_1 I_0 \cos\phi_0 + j V_1 I_0 \sin\phi_0$$

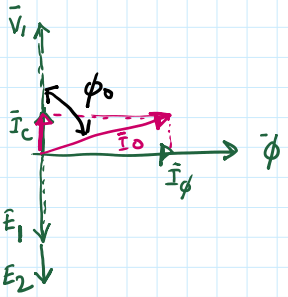
$$= V_1 I_c + j V_1 I_\phi$$

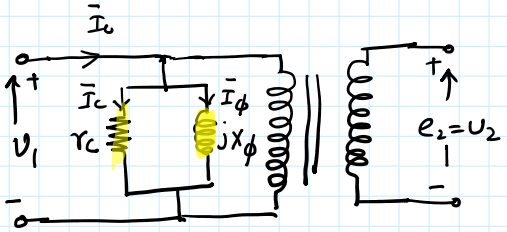
$$\bar{S}_1 = I_c^2 r_c + j I_\phi^2 X_\phi$$

$$= P_1 + j Q_1$$

$$\bar{S}_1 = P_c + j Q_1$$

$$\boxed{P_c = I_c^2 \times r_c}$$

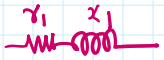




$I_c * r_c = V_1$

$I_\phi X_\phi = V_1$

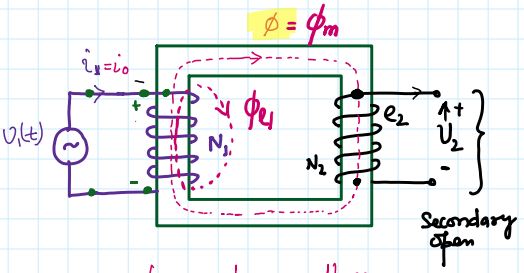
Transformer at No load with Winding Impedance:



r_1 = Resistance of primary winding

x_1 = Leakage reactance of the primary winding

Z_1 = Impedance of primary winding



Transformer at no voltage

Z_1 = Impedance of primary coil
$= r_1 + jX_1$

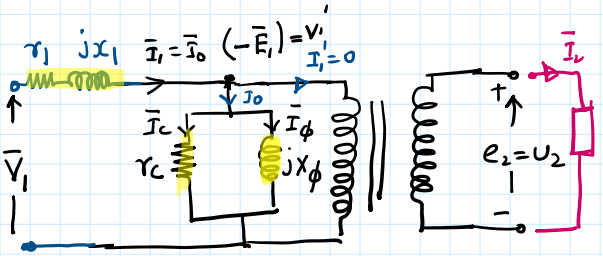
Transformer at no voltage

Secondary open

$\bar{V}_1 = \bar{V}_1' + (r_1 + jX_1)\bar{I}_1$

$\bar{V}_1 = \bar{V}_1' + (r_1 + jX_1)\bar{I}_0$ — (1)

$\bar{V}_1' = -\bar{E}_1$



$\bar{V}_1 = V_1 \angle 0$

$I_c * r_c = V_1$

$I_\phi X_\phi = V_1$

$\bar{S} = \bar{S}_1 = \bar{V}_1 * \bar{I}_0^*$

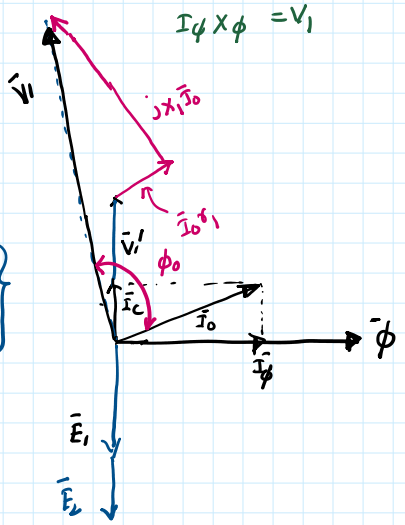
$= \bar{V}_1 * (I_0 \angle -\phi_0)^*$

$= V_1 I_0 \cos\phi_0 + jV_1 I_0 \sin\phi_0$

$= P_1 + jQ_1$

$P_1 = V_1 I_0 \cos\phi_0 = I_0^2 r_1 + I_c^2 r_c$

$Q_1 = V_1 I_0 \sin\phi = I_0^2 X_1 + I_\phi^2 X_\phi$



Losses

Fixed loss = No load loss = Core loss
$P_c = P_h + P_e$
$= I_c^2 r_c$

Variable losses = Copper losses
$I_0^2 r_1 + I_c^2 r_c$; $I_0^2 * X_1 + I_\phi^2 X_\phi$

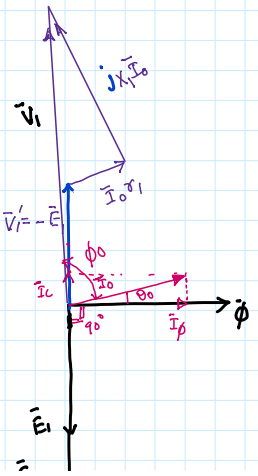
1. Ideal transformer
2. Transformer at no load

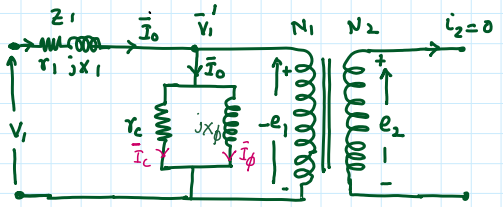
The leakage of flux is accounted as voltage drop with the reactive element $x_{e_1} = x_e$. The reactance x_{e_1} is denoted as leakage reactance of the primary winding. Now with this the phasor diagram is shown below.

$\bar{S}_1 = \bar{V}_1 \bar{I}_0^* = V_1 (I_0 \angle -\phi_0)^* = V_1 I_0 \cos\phi_0 + jV_1 I_0 \sin\phi_0$
$= P_1 + jQ_1$

$P_1 = V_1 I_0 \cos\phi_0 = I_1^2 r_1 + P_c$

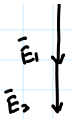
$Q_1 = V_1 I_0 \sin\phi_0 = I_0^2 X_1 + I_\phi^2 X_\phi$





The voltages and currents are related as

$\bar{V}_1 = (-\bar{E}_1) + \bar{I}_0(r_1 + jX_1)$



Phasor diagram at no load with winding resistance

The voltages and currents are related as

$$\begin{cases} \bar{V}_1 = (-\bar{E}_1) + \bar{I}_0(r_1 + jx_1) \\ \bar{V}_1 = \bar{V}_1' + \bar{I}_0(r_1 + jx_1) \\ \bar{V}_2 = \bar{E}_2 \end{cases}$$